OriginChain

AI-Powered Blockchain Product Authentication

Software Requirements Specification

Version 1.0

Confidential — Final Year Engineering Project

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of OriginChain — an AI-powered, blockchain-based product authentication system designed to combat counterfeit products throughout the supply chain.

This document serves as the primary reference for the development team, project guide, evaluators, and future contributors by clearly describing the system's objectives, features, architecture, constraints, and expected behavior.

The SRS establishes a common understanding among all stakeholders and provides the foundation for system design, implementation, testing, deployment, and maintenance.

1.2 Scope

Counterfeit products are a major challenge across industries, causing financial losses, damaging brand reputation, and reducing consumer trust. Traditional product authentication methods such as static barcodes and printed labels can be duplicated or manipulated, making them insufficient for ensuring product authenticity.

OriginChain addresses this challenge by combining blockchain technology, artificial intelligence, and barcode-based verification into a unified authentication platform. The system enables manufacturers to register products and generate unique barcodes linked to blockchain records. As products move through the supply chain — from manufacturer to distributor and retailer — ownership transfers are securely recorded. Consumers can scan the barcode to verify product authenticity and view its history. Additionally, an AI module analyses uploaded product packaging images to detect potential counterfeit indicators, providing an extra layer of verification.

The project focuses on demonstrating a secure, transparent, and scalable approach to product authentication suitable for academic research and future real-world enhancement.

1.3 Objectives

The primary objectives of OriginChain are:

- To reduce counterfeit products using blockchain-based authentication.
- To provide transparent product traceability throughout the supply chain.
- To generate secure and unique barcodes for product verification.
- To maintain immutable ownership records using blockchain technology.
- To assist consumers in verifying product authenticity.
- To integrate AI-based packaging verification for additional counterfeit detection.
- To provide an intuitive web application for all stakeholders.
- To demonstrate the integration of AI, blockchain, and modern web technologies within a single system.

1.4 Definitions, Acronyms and Abbreviations

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token used for authentication
IPFS	InterPlanetary File System for decentralised file storage
Smart Contract	Self-executing blockchain program
Blockchain	Distributed immutable ledger
FastAPI	Python web framework for backend APIs
PostgreSQL	Relational Database Management System
CNN	Convolutional Neural Network
OpenCV	Open Source Computer Vision Library
PyTorch	Deep Learning Framework
SKU	Stock Keeping Unit
UUID	Universally Unique Identifier

1.5 References

The following resources were considered during planning and design of OriginChain:

- IEEE Software Requirements Specification (IEEE 830 / ISO/IEC/IEEE 29148)
- FastAPI Official Documentation
- Next.js Official Documentation
- React Official Documentation
- PostgreSQL Documentation
- Solidity Documentation
- Polygon Documentation
- Hardhat Documentation
- IPFS Documentation
- OpenCV Documentation
- PyTorch Documentation

1.6 Document Overview

This document is organised into sections describing the overall system, user roles, functional and non-functional requirements, interfaces, architecture, database overview, use cases, and future enhancements. It is intended to guide the complete software development lifecycle of OriginChain — from planning through deployment and maintenance.

2. Overall Description

2.1 Product Perspective

OriginChain is a web-based product authentication and traceability system designed to reduce counterfeit products by combining blockchain technology, artificial intelligence, and barcode verification.

The system provides a secure platform where manufacturers can register products and generate unique barcodes linked to blockchain records. As products move through the supply chain, ownership transfers are recorded, creating an immutable history of the product's journey.

Consumers can verify the authenticity of products by scanning the barcode using the web application. The system retrieves product information, verifies blockchain records, and optionally allows consumers to upload an image of the product packaging for AI-based authenticity analysis.

OriginChain integrates the following technologies into a single platform:

- Frontend: Next.js, React, Tailwind CSS
- Backend: FastAPI, Python
- Database: PostgreSQL
- Blockchain: Solidity, Polygon, Hardhat
- Decentralised Storage: IPFS
- Artificial Intelligence: OpenCV and PyTorch

2.2 Product Functions

OriginChain provides the following core functionalities:

Authentication & Authorisation

- Secure user login using JWT authentication.
- Role-based access control for Admin, Manufacturer, Distributor, and Retailer.
- Consumer access to verification features without requiring an account.

Product Registration

- Manufacturers can register new products.
- Generate unique Product IDs.
- Generate unique QR codes for each registered product.
- Store product information in the database.
- Store authenticity records on the blockchain.

Supply Chain Management

- Transfer product ownership from Manufacturer to Distributor.
- Transfer product ownership from Distributor to Retailer.
- Record every ownership transfer on the blockchain.
- Maintain a complete ownership history.

Product Verification

- Consumers scan barcodes.
- Retrieve product information.
- Verify authenticity using blockchain records.
- Display product details and ownership history.

AI-Based Verification

- Upload product packaging images.
- Analyse packaging using a trained AI model.
- Detect possible counterfeit indicators.
- Display AI prediction and confidence score.

Administration

- Manage users.
- View registered products.
- Monitor verification activities.
- Monitor blockchain transactions.
- View AI verification logs.

2.3 User Classes and Characteristics

Admin

The Administrator manages the entire platform, including user management, product oversight, verification monitoring, blockchain activity monitoring, and system settings. Technical knowledge required: Moderate.

Manufacturer

Manufacturers create and manage products within the system. Responsibilities include registering products, generating barcodes, uploading product details, and initiating ownership transfers. Technical knowledge required: Basic to Moderate.

Distributor

Distributors receive products from manufacturers and transfer them to retailers. They accept and transfer ownership and view product history. Technical knowledge required: Basic.

Retailer

Retailers receive products and prepare them for sale. They accept ownership, verify product details, and view ownership history. Technical knowledge required: Basic.

Consumer

Consumers verify products before or after purchase by scanning barcodes, viewing product information, and optionally uploading product packaging images for AI verification. Technical knowledge required: Minimal.

2.4 Operating Environment

Client Side

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari (basic compatibility)

Server Side

- Ubuntu Linux (recommended)
- Windows 11 (development)
- Python 3.12+, FastAPI, Uvicorn

Database

- PostgreSQL 16+

Blockchain

- Polygon Amoy Testnet (development)
- Polygon Mainnet (future deployment)

Development Tools

- Visual Studio Code, Git, GitHub
- Postman, pgAdmin
- Hardhat, MetaMask

2.5 Design Constraints

- The project is developed as a final-year engineering project with a team of four members.
- The system is initially designed as a web application.
- Blockchain interactions will use the Polygon network.
- Smart contracts will be developed using Solidity.
- PostgreSQL will serve as the primary relational database.
- AI verification will focus on packaging image analysis.
- The project timeline is limited by the academic schedule.
- Development will prioritise modularity and maintainability.

2.6 Assumptions and Dependencies

Assumptions

- Users have internet connectivity.
- Manufacturers provide accurate product information.
- QR codes remain attached to products throughout the supply chain.
- Blockchain network is available when transactions are performed.

- AI model is trained using an appropriate dataset before deployment.

Dependencies

- FastAPI, PostgreSQL, SQLAlchemy, JWT libraries
- Polygon blockchain, Solidity smart contracts, Hardhat
- IPFS, OpenCV, PyTorch
- Barcode generation library

3. Functional Requirements

3.1 Authentication Module

The Authentication Module is responsible for securely identifying users and granting access based on their assigned roles. The system uses JSON Web Tokens (JWT) to authenticate users and Role-Based Access Control (RBAC) to restrict access to authorised functionalities.

FR ID	Requirement
FR-1.1	The system shall allow registered users to log in using their email address and password.
FR-1.2	The system shall validate user credentials before granting access.
FR-1.3	The system shall generate a JWT token upon successful authentication.
FR-1.4	The system shall require a valid JWT token for all protected API requests.
FR-1.5	The system shall restrict access to features based on user roles.
FR-1.6	The system shall deny access to unauthorised users.
FR-1.7	The system shall allow users to log out by invalidating the client session.

3.2 User Management Module

The User Management Module enables administrators to manage users participating in the supply chain.

FR ID	Requirement
FR-2.1	The Administrator shall be able to create user accounts.
FR-2.2	The Administrator shall assign roles during user creation (Manufacturer, Distributor, Retailer, Admin).
FR-2.3	The Administrator shall update user information.
FR-2.4	The Administrator shall activate or deactivate user accounts.
FR-2.5	The Administrator shall view all registered users.
FR-2.6	The system shall store encrypted passwords.

3.3 Product Management Module

Manufacturers use this module to define products that will later be manufactured in batches. A Product represents the generic item (e.g., "Nike Air Max 270") and stores information common to all produced units.

FR ID	Requirement
FR-3.1	The Manufacturer shall create a new Product.
FR-3.2	Each Product shall have a unique Product ID.
FR-3.3	The Manufacturer shall provide: Product Name, Brand, Category, Description, Product Images, and Product Specifications.
FR-3.4	The system shall allow updating product details.
FR-3.5	The system shall maintain product status (Active / Discontinued).

3.4 Product Batch Management Module

A Product Batch represents a manufacturing lot of a product. Each batch contains multiple Product Units.

FR ID	Requirement
FR-4.1	Manufacturers shall create batches for existing products.
FR-4.2	Each batch shall have: Batch Number, Manufacturing Date, Expiry Date (if applicable), and Quantity Produced.
FR-4.3	The system shall associate every batch with exactly one Product.
FR-4.4	The system shall generate Product Units for the specified quantity.
FR-4.5	Each generated Product Unit shall receive a unique identifier.

3.5 Product Unit Management Module

Each Product Unit represents one physical item available in the supply chain.

FR ID	Requirement
FR-5.1	Every Product Unit shall have a unique Product Unit ID.
FR-5.2	Every Product Unit shall have one unique barcode.
FR-5.3	The system shall maintain the current owner of each Product Unit.

FR ID	Requirement
FR-5.4	The system shall maintain the current product status (Manufactured / In Transit / With Distributor / With Retailer / Sold).
FR-5.5	The system shall prevent duplicate Product Unit IDs.

3.6 Barcode Management Module

The Barcode Management Module is responsible for generating and managing unique barcodes for each Product Unit.

FR ID	Requirement
FR-6.1	The system shall generate one Code 128 barcode for every Product Unit.
FR-6.2	Every barcode shall uniquely identify one Product Unit.
FR-6.3	The barcode shall not directly store complete product information.
FR-6.4	Scanning the barcode shall retrieve the associated Product Unit details through the backend.
FR-6.5	The system shall prevent duplicate barcode generation.

3.7 Supply Chain & Ownership Transfer Module

This module tracks the movement of products through the supply chain while maintaining ownership history.

FR ID	Requirement
FR-7.1	Manufacturers shall transfer Product Units to Distributors.
FR-7.2	Distributors shall transfer Product Units to Retailers.
FR-7.3	The system shall record every ownership transfer.
FR-7.4	The blockchain shall store ownership transfer records.
FR-7.5	The system shall maintain complete ownership history for every Product Unit.

3.8 Product Verification Module

Consumers verify the authenticity of a product by scanning its barcode.

FR ID	Requirement
FR-8.1	Consumers shall scan a barcode without creating an account.
FR-8.2	The system shall retrieve Product Unit details.
FR-8.3	The system shall verify the blockchain authenticity record.
FR-8.4	The system shall display: Product Details, Manufacturer, Batch Information, Ownership History, and Verification Result.

3.9 AI Verification Module

Consumers may optionally upload a packaging image for AI-based analysis.

FR ID	Requirement
FR-9.1	The system shall accept product packaging images.
FR-9.2	The AI model shall analyse the uploaded image.
FR-9.3	The system shall return: Prediction and Confidence Score.
FR-9.4	The AI verification result shall be linked to the corresponding Product Unit.

3.10 Blockchain Module

The Blockchain Module ensures authenticity and traceability through immutable records.

FR ID	Requirement
FR-10.1	The system shall register authenticity records on the blockchain.
FR-10.2	The system shall record ownership transfers on the blockchain.
FR-10.3	The system shall verify blockchain records during product verification.

FR ID	Requirement
FR-10.4	The system shall store blockchain transaction references in PostgreSQL.

3.11 Dashboard Module

The system shall provide dashboards appropriate to each authenticated role:

- Admin Dashboard: System overview, users, products, verification logs.
- Manufacturer Dashboard: Products, batches, units, ownership transfers initiated.
- Distributor Dashboard: Received inventory, transferred inventory.
- Retailer Dashboard: Current inventory, verification status before sale.

3.12 Audit & Verification Logs

FR ID	Requirement
FR-12.1	The system shall maintain logs of all product verification requests.
FR-12.2	The system shall record: Verification date and time, Product Unit ID, Verification outcome, AI verification result (if performed).
FR-12.3	The Administrator shall be able to view verification logs.

4. External Interface Requirements

4.1 User Interfaces

OriginChain is a web-based application accessible through modern web browsers. The interface is designed to be responsive, user-friendly, and role-specific. Each user role is presented with a dedicated dashboard containing features relevant to their responsibilities.

4.1.1 Login Interface

The Login Interface shall allow registered users (Admin, Manufacturer, Distributor, and Retailer) to securely authenticate using their email address and password. The interface shall include:

- Email field and Password field
- Show/Hide Password option
- Login button
- Validation messages for incorrect credentials

4.1.2 Admin Dashboard

The Admin Dashboard shall provide a system-wide overview including: User Management, Product and Batch Overview, Product Unit Overview, Verification Logs, AI Verification Logs, Blockchain Transaction Monitoring, and Search and Filter Options.

4.1.3 Manufacturer Dashboard

The Manufacturer Dashboard shall allow manufacturers to: Create Products, Create Product Batches, Generate Product Units and Barcodes, View Inventory, Initiate Ownership Transfer, and View Product History.

4.1.4 Distributor Dashboard

The Distributor Dashboard shall allow distributors to: View Received Products, Accept Ownership, Transfer Ownership to Retailers, View Product History, and Search Products.

4.1.5 Retailer Dashboard

The Retailer Dashboard shall allow retailers to: View Received Products, Verify Product Authenticity, and View Product Details and Ownership History.

4.1.6 Consumer Verification Interface

Consumers are not required to log in. The Consumer Verification page shall allow users to: enter or scan the product barcode, view product details and authenticity status, view ownership history, upload packaging images, and view AI verification results.

4.2 Hardware Interfaces

The system requires client devices (desktop, laptop, or smartphone) with an internet connection and a modern web browser. Optionally, a camera may be used for barcode scanning and packaging image upload.

4.3 Software Interfaces

Frontend

- Next.js, React, Tailwind CSS

Backend

- FastAPI, SQLAlchemy, JWT Authentication

Database

- PostgreSQL

Blockchain

- Solidity Smart Contracts, Polygon Network, Hardhat, MetaMask

AI

- OpenCV, PyTorch, Isolation Forest

External Libraries

- Barcode Generation Library, Pillow (Image Processing), OpenCV, Web3.py

4.4 Communication Interfaces

Communication between system components shall occur using REST APIs over HTTPS. Data exchanged between the frontend and backend shall use JSON format. Blockchain communication shall occur through Web3-compatible interfaces.

5. Non-Functional Requirements

5.1 Performance

The system should respond to standard API requests within approximately 2 seconds under normal conditions, handle concurrent requests from multiple authenticated users during demonstrations, and generate barcodes efficiently after Product Unit creation.

5.2 Security

The system shall encrypt passwords before storage, use JWT authentication and Role-Based Access Control, validate all user inputs, protect APIs from unauthorised access, and use HTTPS in deployment.

5.3 Reliability

The system should maintain data consistency between PostgreSQL and blockchain records, prevent duplicate Product Units and barcode generation, and record failed verification attempts where applicable.

5.4 Availability

The system should remain available whenever the server and blockchain network are operational. If blockchain services are temporarily unavailable, the system should notify the user appropriately instead of failing silently.

5.5 Scalability

The architecture should support additional manufacturers, distributors, retailers, increased product batches, and increased verification requests without requiring significant architectural changes.

5.6 Maintainability

The software shall be developed using a modular architecture. Each module (Frontend, Backend, Blockchain, AI) shall be independently maintainable. Source code shall be documented and organised using consistent coding standards.

5.7 Usability

The system shall provide simple navigation, clear error messages, a responsive design, a minimal learning curve, and a consistent user interface across all dashboards.

5.8 Portability

The web application shall operate on modern browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

6. System Architecture

OriginChain follows a six-layer architecture where each layer has a distinct responsibility:

Layer	Primary Responsibility
Presentation Layer	User interaction — Next.js, React, Tailwind CSS
Application Layer	API routing, authentication, input validation — FastAPI
Business Logic Layer	Product registration, batch creation, unit generation, ownership transfer
Blockchain Layer	Immutable authenticity and transfer records — Solidity, Polygon, Hardhat
AI Layer	Image preprocessing, packaging verification, counterfeit prediction — OpenCV, PyTorch
Data Layer	Structured data storage — PostgreSQL, IPFS

7. Database Overview

The system uses PostgreSQL as its primary relational database. The major entities are:

- User
- Product
- Product Batch
- Product Unit
- Ownership Transfer
- Verification Log
- AI Verification
- Blockchain Transaction

Each Product can have multiple Product Batches. Each Product Batch can contain multiple Product Units. Each Product Unit has one unique barcode and one current owner. Ownership Transfer records maintain the movement of Product Units throughout the supply chain. Verification Logs store every barcode verification request. AI Verification records store packaging image analysis results. Blockchain Transaction records maintain references to blockchain transactions associated with Product Units.

8. Use Cases

The primary use cases supported by the system are:

- User Login
- Register Product
- Create Product Batch
- Generate Product Units
- Generate Barcode
- Transfer Ownership
- Verify Product Authenticity
- Upload Packaging Image
- AI Verification
- View Product History
- Manage Users
- View Verification Logs

Detailed use case specifications (actors, preconditions, main flow, alternate flows, and postconditions) can be added in a later revision.

9. Future Scope

Potential enhancements for future versions of OriginChain include:

- Native Android and iOS applications
- IoT-based product tracking
- NFC tag integration
- Push notifications
- Advanced analytics dashboards
- Multi-language support
- ERP integration
- GS1 barcode compatibility
- AI model improvements with larger datasets

10. References

- IEEE 29148: Systems and Software Engineering — Life Cycle Processes — Requirements Engineering
- FastAPI Official Documentation
- React Official Documentation
- Next.js Official Documentation
- PostgreSQL Documentation
- SQLAlchemy Documentation
- Solidity Documentation
- Polygon Documentation
- Hardhat Documentation
- IPFS Documentation
- OpenCV Documentation
- PyTorch Documentation